

STUDENT PERFORMANCE ANALYSIS

Python Project

Using Python, NumPy,
Pandas, Matplotlib and
Seaborn

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Objectives

- **Understand the dataset.**
- **Load CSV using Pandas.**
- **Perform data cleaning.**
- **Understand dataset structure.**
- **Analyze student performance.**
- **Perform statistical analysis.**
- **Create visualizations.**

Phase 1 - Dataset Loading

In this phase, students are required to import the necessary Python libraries, load the CSV dataset into a Pandas DataFrame, and verify that the dataset has been loaded successfully.

Question 1: Import the required Python libraries (NumPy, Pandas, Matplotlib and Seaborn).

Question 2: Load the CSV dataset into a Pandas DataFrame.

Question 3: Store the dataset inside a variable named 'df'.

Question 4: Display the dataset and verify that it has been loaded successfully.

Phase 2 - Dataset Understanding

Before performing any analysis, it is important to understand the structure, contents, and characteristics of the dataset. Complete the following tasks.

Question 5: Display the first five records of the dataset using an appropriate Pandas function.

Question 6: Display the last five records of the dataset.

Question 7: Find the shape of the dataset and identify the total number of rows and columns.

Question 8: Display all column names present in the dataset.

Question 9: Display the data type of every column.

Question 10: Display complete information about the dataset.

Question 11: Generate the statistical summary of all numerical columns.

Question 12: Display five random records from the dataset.

Question 13: Identify all numerical columns in the dataset.

Question 14: Identify all categorical columns in the dataset.

Phase 3 - Data Cleaning

Data cleaning is an important step before performing data analysis. Students must verify the quality of the dataset and ensure that it is free from missing values, duplicate records and data inconsistencies.

Question 15: Check whether the dataset contains any missing (NULL) values.

Question 16: Check whether the dataset contains duplicate records.

Question 17: Verify the data types of every column in the dataset.

Question 18: Verify the shape of the dataset after completing the cleaning process.

Question 19: Save the cleaned dataset as a new CSV file.

Phase 4 - Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is performed to understand the dataset through statistical summaries and meaningful observations. Answer the following analytical questions using Python, NumPy and Pandas.

Question 20: Find the total number of students available in the dataset.

Question 21: Calculate the average Attendance and Average CGPA.

Question 22: Find the highest and lowest Total Marks obtained by students.

Question 23: Perform Department-wise Average CGPA Analysis.

Question 24: Perform Gender-wise Performance Analysis.

Question 25: Identify the Top 10 students based on CGPA.

Question 26: Calculate the correlation between all numerical columns.

Phase 5 - Data Visualization

Data visualization helps transform numerical information into meaningful charts and graphs, making it easier to identify trends, patterns, and relationships within the dataset.

Question 27: Create suitable visualizations (Bar Chart, Histogram, Pie Chart, Scatter Plot and Heatmap) and explain the insights obtained.

Question 28: Compare Attendance vs CGPA and Study Hours vs Total Marks using appropriate graphs.

Question 29: Write the overall findings and final conclusion based on your analysis.
